Coupled Forward ECG on Real-Geometry FEM: ASM vs. sASM Domain-Decomposition Preconditioning

Branch `cardiac-sim-fakegeo` · algorithm study (numerical linear algebra / domain decomposition)

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MFEM 4.9 + PETSc 3.19 · all numbers measured on `heart.msh`

Overview: nature of the study and the main thread

Nature: a Krylov-preconditioning / domain-decomposition (Additive Schwarz) *algorithm* study. The vehicle is a coupled-PDE forward problem, but the *object of study is how the linear systems converge faster and scale*.

Pipeline

$$\underbrace{\text{Sys1 } V_m}_{\text{parabolic}\cdot\text{local}} \xrightarrow{V_m(t)} \underbrace{\text{Sys2 } u_e}_{\text{singular elliptic}\cdot\text{global}} \xrightarrow{\text{Dirichlet}} \underbrace{\text{Sys3 } \varphi}_{\text{elliptic}\cdot\text{nonsingular}} \rightarrow \text{ECG}$$

This deck focuses on: the problem definition, the mesh, and the **ASM(BASIC)** vs. **sASM(scaled/restricted) iteration counts** and **wall-clock time** on the three systems.

Problem: algebraic characterization of the three systems

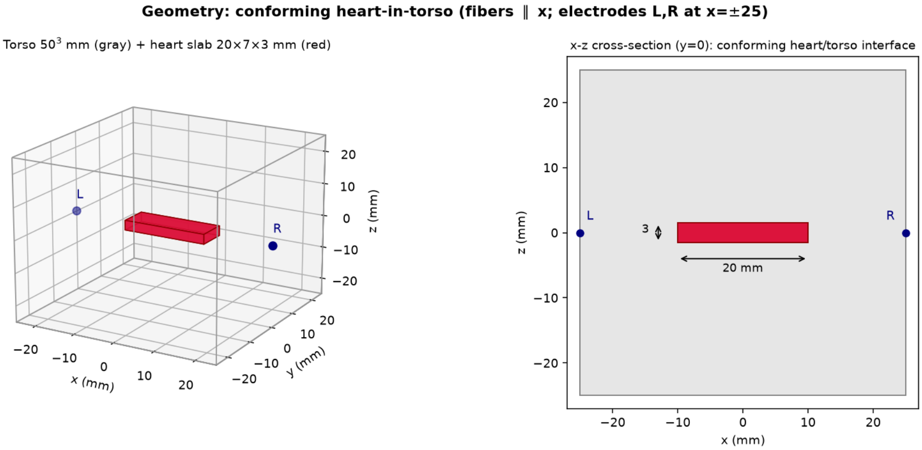
Each time step solves three SPD (semi-)definite systems (`heart.msh`, Sys2 has 10085 unknowns):

System	Equation	Algebraic type	baseline iters
Sys1 V_m	$\left(\frac{1}{\Delta t}M + \frac{1}{2}K\right) V_m = b$	parabolic / **mass-dominated** / nonsingular	CG+ICC $\sim$13
Sys2 u_e	$K_{\sigma_i+\sigma_e}\, u_e = -K_{\sigma_i} V_m$	elliptic / **pure-Neumann singular** / global	$\sim$82–96
Sys3 φ	$K_{\sigma_o}\varphi = 0$, interface Dirichlet	elliptic / nonsingular / global	$\sim$62–66

Sys2 is the only hard one (singular + elliptic + global):

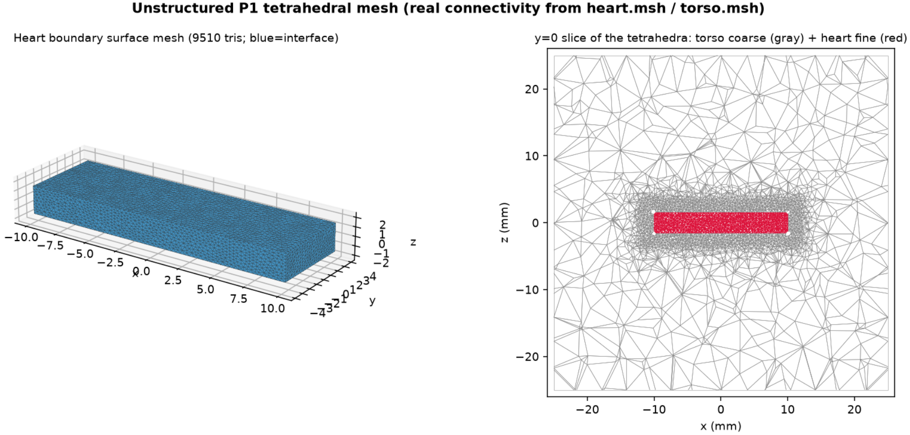
- $K\mathbf{1} = 0$: null space = constants = one slow global mode $\Rightarrow$ de-mean the RHS + `MatSetNullSpace` (project the constant mode out every CG iteration).
- global elliptic $\Rightarrow$ a one-level domain decomposition propagates only one subdomain hop per iteration; more subdomains = slower (the core theme below).

Geometry: conforming heart-in-torso domains



Heart slab $20 \times 7 \times 3$ mm centered in a 50^3 mm torso; **conforming** interface (Gmsh `BooleanFragments` $\Rightarrow$ shared interface surface). Anisotropic σ, fibers $\parallel$ x; electrodes L,R at $x = \pm 25$; subdomains = MPI ranks (METIS).

Mesh: unstructured P1 tetrahedra (real connectivity)



heart	10 085 nodes	47 582 tets	9 510 tris
torso	36 229 nodes	206 647 tets	12 112 tris

interface shared dofs: 4 757; y=0 slice: heart ∼0.5 mm, torso ∼3 mm

Method: ASM(BASIC) vs. sASM(scaled / restricted)

One-level additive Schwarz, blocks = METIS subdomains, sub-solve = ICC(0):

$$M_{\mathrm{ASM}}^{-1} = \sum_i R_i^\top A_i^{-1} R_i, \qquad A_i = R_i A R_i^\top \text{ (overlap } O \text{ layers)}$$

The key difference — overlap *multiplicity* (over-count)

- **ASM(BASIC)**: overlap dofs get *one contribution from each* overlapping subdomain $\Rightarrow$ **over-counting** in the overlap, which corrupts the operator spectrum and can even *increase iterations with overlap*.
- **sASM**: a diagonal partition of unity $D^{-1/2}(\cdot)D^{-1/2}$ cancels the over-count ($D =$ multiplicity) $\Rightarrow$ overlap works *as intended*, iterations drop with overlap.

$$M_{\mathrm{sASM}}^{-1} = D^{-1/2}\left(\sum_i R_i^\top A_i^{-1} R_i\right) D^{-1/2}$$

At O=0 (no overlap) there is no over-count, so **ASM** $\equiv$ **sASM**.

Iterations: ASM vs. sASM (n_p=4, rtol 10^{-8}, ICC(0))

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	11	11	O=0: equal
	1	13 ↑	**8** ↓	
	2	13 ↑	**8** ↓	
Sys2 u_e (singular)	0	82	82	
	1	92 ↑	**74** ↓	
	2	93 ↑	**74** ↓	
Sys3 φ	0	62	62	
	1	88 ↑	**53** ↓	
	2	94 ↑	**53** ↓	

↑ ASM iterations *rise* with overlap (over-count); ↓ sASM *falls* with overlap (overlap works correctly).

Iterations: ASM vs. sASM (n_p=8, more subdomains $\Rightarrow$ stronger anomaly)

System	O	ASM(BASIC)	sASM	note
Sys1 V_m	0	13	13	
	1	18 $\uparrow$	**8** $\downarrow$	
	2	19 $\uparrow$	**8** $\downarrow$	
Sys2 u_e (singular)	0	79	79	
	1	117 $\uparrow$	**73** $\downarrow$	
	2	123 $\uparrow$	**71** $\downarrow$	
Sys3 φ	0	66	66	
	1	93 $\uparrow$	**55** $\downarrow$	
	2	109 $\uparrow$	**53** $\downarrow$	

Most extreme: Sys3 ASM O:0$\rightarrow$2 66$\rightarrow$93$\rightarrow$109 (+65%); sASM 66$\rightarrow$55$\rightarrow$53 ($-$20%).

Wall-clock: ASM vs. sASM (ms; KSP+PCASM setup + solve, avg of 20)

$n_p = 4$

System	O	ASM	sASM
Sys1	0	3.66	3.79
	1	5.27	**5.60**
	2	6.88	**5.22**$\downarrow$
Sys2	0	15.39	18.85
	1	21.50	**17.60**$\downarrow$
	2	24.33	**19.39**$\downarrow$
Sys3	0	43.59	46.99
	1	68.10	**44.36**$\downarrow$
	2	80.18	**52.57**$\downarrow$

$n_p = 8$

System	O	ASM	sASM
Sys1	0	5.37	5.62
	1	8.93	**7.89**$\downarrow$
	2	14.90	**8.91**$\downarrow$
Sys2	0	24.48	26.73
	1	45.72	**28.94**$\downarrow$
	2	53.15	**32.83**$\downarrow$
Sys3	0	72.98	59.58$\downarrow$
	1	96.81	**71.03**$\downarrow$
	2	133.24	**77.17**$\downarrow$

For $O \geq 1$, sASM is **both fewer iterations and faster wall-clock**; for ASM, overlap *costs time and raises iterations*. $n_p = 8$ Sys3 $O = 2$: ASM 109it/133ms vs. sASM 53it/77ms ($\sim 1.7\times$).

Key finding: the overlap anomaly

Observation

Adding overlap to one-level **ASM(BASIC)** *raises* iterations and slows wall-clock; adding overlap to **sASM** *lowers both*.

Mechanism: overlap anomaly = over-count × inexactness

- overlap dofs counted once per subdomain ⇒ BASIC **amplifies** in the overlap ($\|M^{-1}\|$ too large), degrading the spectrum.
- ICC(0) sub-solves are *inexact*, so the amplified error is not absorbed by an exact local solve ⇒ iterations climb.
- the $D^{-1/2}$ partition of unity in sASM cancels the over-count exactly ⇒ restores the correct behavior "more overlap = faster information transport".

Conclusion: on real-geometry FEM, **scaled/restricted ASM is the correct way to use overlapping DD**; BASIC overlap is harmful here. More subdomains (n_p:4→8) ⇒ stronger anomaly, underscoring that *scalability* needs the correct weighting (and, later, a coarse space).

Summary and outlook

- **Problem**: coupled forward ECG on three systems; Sys2 (singular pure-Neumann elliptic) is the only hard one and the target of the preconditioning study.
- **Mesh**: conforming unstructured tet P1-FEM, heart 10085 dof / 47582 tets, torso 36229 / 206647.
- **ASM vs. sASM** (measured, $n_p{=}4, 8$):
  - ASM(BASIC) iterations rise with overlap (over-count), wall-clock slower;
  - sASM iterations fall with overlap, and wall-clock is faster for $O{\geq}1$;
  - at $O{=}0$ the two coincide.
- **Next steps** (implemented / planned on the same branch): strong fine level **SORAS** (84→20, 4.2×), shared **Nicolaides coarse space** (scalability), cross-time **Fischer** (−12%); at scale (∼3000 cores) a **GenEO spectral coarse space** + a telescoped scalable coarse solve.

All numbers measured in-container (MFEM 4.9 + PETSc 3.19); ECG bit-identical (verified).